PYTHON MINI PROJECT

A Stock Market Data Analyzer in Python

One Python file that reads live NSE prices and reports return, risk and trend.

Dax Varasadiya
Enrollment No. 12402080601026

Division IT-A
Department of Information Technology

Stock Market Data Analyzer

OBJECTIVES

Four things the tool has to do

01 **Read any Indian share**

A ticker typed directly, or a company name searched live.

02 **Show the numbers that matter**

Return, volatility, drawdown and a trend signal.

03 **Work at any timeframe**

One-minute candles through five-year weekly bars.

04 **Run without setup**

No API key, no database, no third-party package.

CONCEPTS

Four syllabus concepts, one file

Concept	How it is used	Where in the program
Lists	Dates, closing prices and volumes held as parallel lists	fetch_stock, simple_moving_average
Dictionaries	Every computed statistic returned as one labelled dictionary	analyse, search_symbols
File handling	A price series written to CSV and read back with the csv module	save_csv, load_csv
GUI	Tkinter window with form, table, chart and indices box	launch_gui

ARCHITECTURE

How a request moves through the system

01 Symbol
A ticker typed, or a company name searched.

02 Download
Yahoo Finance returns daily or intraday bars.

03 Analyse
Returns, moving averages, volatility, drawdown.

04 Render
Report, chart and indices in the window.

Stage 03 is the only stage that does arithmetic. The other three move data or draw it. Nineteen functions, one file.

DATA SOURCE

Live prices without an API key

01 **No key, no signup**

Yahoo's public chart endpoint answers without registration.

02 **NSE and BSE only**

Symbols ending .NS and .BO; foreign listings are filtered out.

03 **Minute bars to weekly bars**

The same endpoint serves every timeframe the tool offers.

04 **Exchange time, not machine time**

Intraday stamps read 09:15 to 15:30 for the NSE, whatever the laptop's clock says.

MEASURES

One year of Reliance, in four figures

−19.3%

TOTAL RETURN

₹1,484 to ₹1,198 over the year.

21.2%

VOLATILITY

Annualised from daily returns.

−27.2%

MAX DRAWDOWN

January peak to the October low.

247

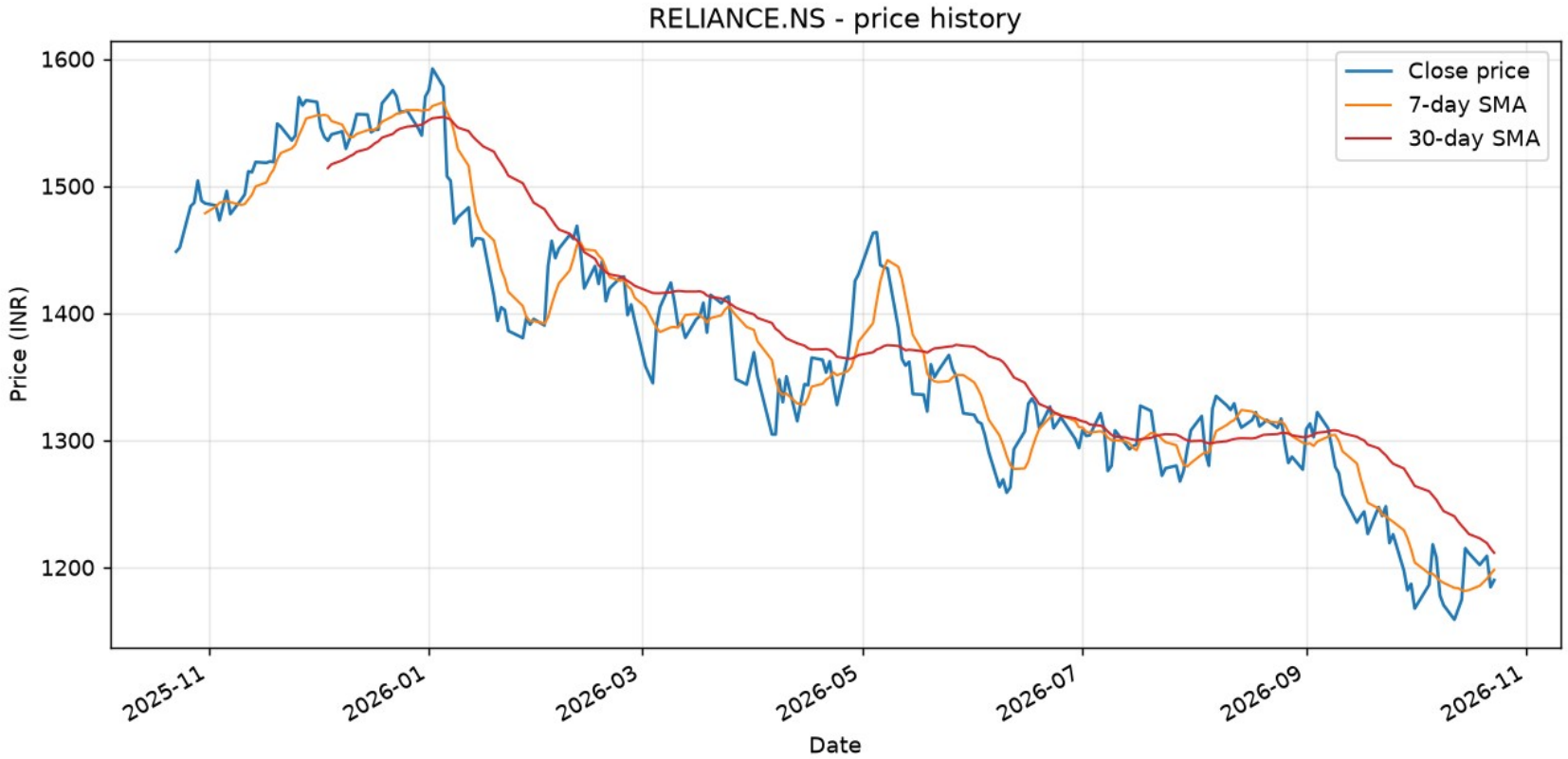
TRADING DAYS

Daily closes in the series.

Source: Yahoo Finance · RELIANCE.NS · 27 Oct 2025 – 26 Oct 2026

TREND

The line falls after January



RELIANCE.NS, one year of daily closes with the 7-day and 30-day moving averages

TIMEFRAMES

Intraday views and market indices

01 One minute

A single session, 375 bars from 09:15 to 15:30.

02 Five minutes

75 bars a day; the axis labels every gap in minutes.

03 One hour

A month of sessions, dated on the axis.

04 Indices box

Nifty 50, Sensex and Bank Nifty on one line, refreshed on demand.

COVERAGE

Six timeframes to choose

Timeframe	Range · interval	Bars	Register
1 day, 1-minute	1d · 1m	375	Intraday
1 day, 5-minute	1d · 5m	75	Intraday
1 week, 15-minute	5d · 15m	125	Intraday
1 month, 1-hour	1mo · 60m	138	Intraday
1 year, daily	1y · 1d	247	Daily
5 years, weekly	5y · 1wk	262	Weekly

Bar counts are for RELIANCE.NS and TCS.NS on 26 October 2026; the exact number varies by share and date.

SEARCH

Any Indian share, by name

01 Type the company, not the ticker

"tata motors" returns TMCV.NS and TMPV.NS, the two halves after the split.

02 Indian listings only

Symbols ending .NS or .BO survive; NYSE, Nasdaq and other exchanges are dropped.

03 Double-click to load

The chosen row fills the symbol box and runs the analysis at once.

04 A miss is explained, not crashed

"No Indian shares found" replaces a traceback when a name has no listing.

VERIFICATION

Tested on live market data

Check	Result	What was observed
Live daily fetch	Pass	RELIANCE.NS returned 247 daily bars for one year.
Intraday fetch	Pass	1-day 5-minute returned 75 bars, labelled 09:15 to 15:25 IST.
Wrong symbol	Pass	A clear "symbol not found" message, no traceback.
Offline demo	Pass	Full report and text chart with no network connection.
CSV round trip	Partial	Prices reload; the interval is not stored, so it reads as daily.

The CSV gap is the one known defect: saving the interval alongside the prices would close it.

LIMITS

Known limits and next steps

01 Live mode needs a connection

Without internet only the generated demonstration data is available.

02 One data source

Everything depends on a single public endpoint that could change.

03 Indicators are simple

Two moving averages and volatility; no RSI, MACD or backtesting.

04 No database

Prices live in CSV files, so there is no history, watchlist or SQL.

Thank You